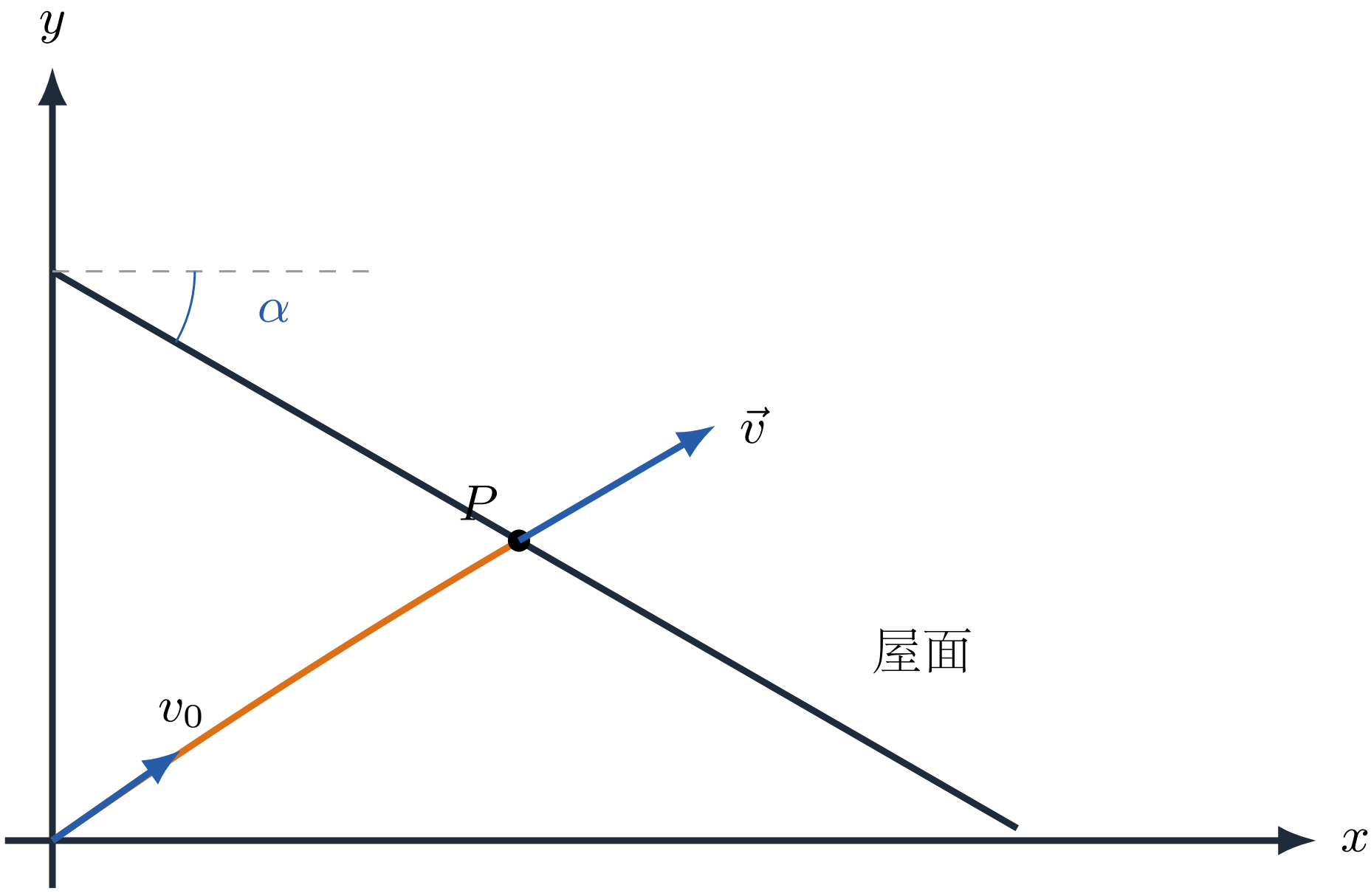

$$\theta_{\text{opt}} = \pi/2 - \alpha$$

$$y = h - x\tan\alpha$$